



University of East London

**SCHOOL OF ARCHITECTURE, COMPUTING &
ENGINEERING**

INDIVIDUAL CW REPORT

TOPIC: DIGITAL CRIME SCENE INVESTIGATION

Student Name: Tejakumar GOVINDARAJULU

Student Number: 2855167

Module Leader: Dr. Tauseef Ahmed

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Digital Forensic Investigation

1. Introduction

Digital forensics has been a powerful tool for criminal investigations, as it sheds light on user interaction, data processing, and other hidden facets. With the emergence of the digital black market exploiting Autopsy 4.22.1's capabilities, this tool plays an even more important role as a forensic weapon. This report will present a forensic analysis of the case with suspicion of cyber activity regarding the Snowy Owl (*Bubo scandi* us), an animal protected by international publications on conservation of fauna and flora.

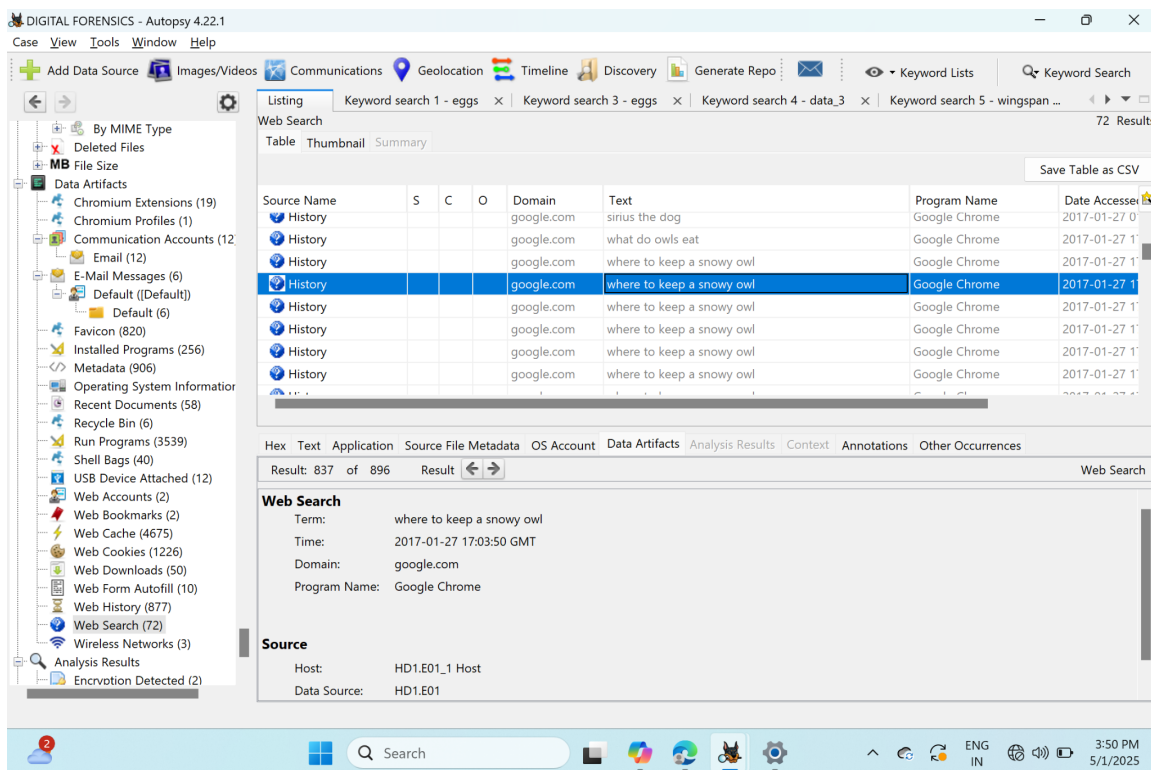
This report will summarize the results of a digital forensic inspection conducted during an investigation on whether a person, with the nickname of Sarah McAvoy, had any suspicious cyber activity related to Snowy Owl. We utilized e-mail accounts, systems metadata, web search histories, internet communication services such as Skype, log files, and even deleted files produced by the Chrome web browser. The most disturbing results were the “where do you keep snowy owls” phrases and “deleted_image_owl_85.jpeg” images and suspicious email masks registered in disguised domains.

Opening an archived remote image during the examination confirmed the hypotheses that there are files that had been deleted and even some that were deliberately altered—selection of the copyrighted materials, a significant amount of marketing material, and browsing of the wildlife site. It also contains private communication having a password in which a user, whose name can be given accordingly, is inviting a friend for a kind of competitive tournament, who quotes fables, and it goes: I will bring you to the contest. From the examination conducted, it was not possible to obtain any information that would indisputably prove the existence of a transaction in the illegal trade of wildlife. But it could reasonably be considered the creature of the set of digital evidence, albeit circumstantial.

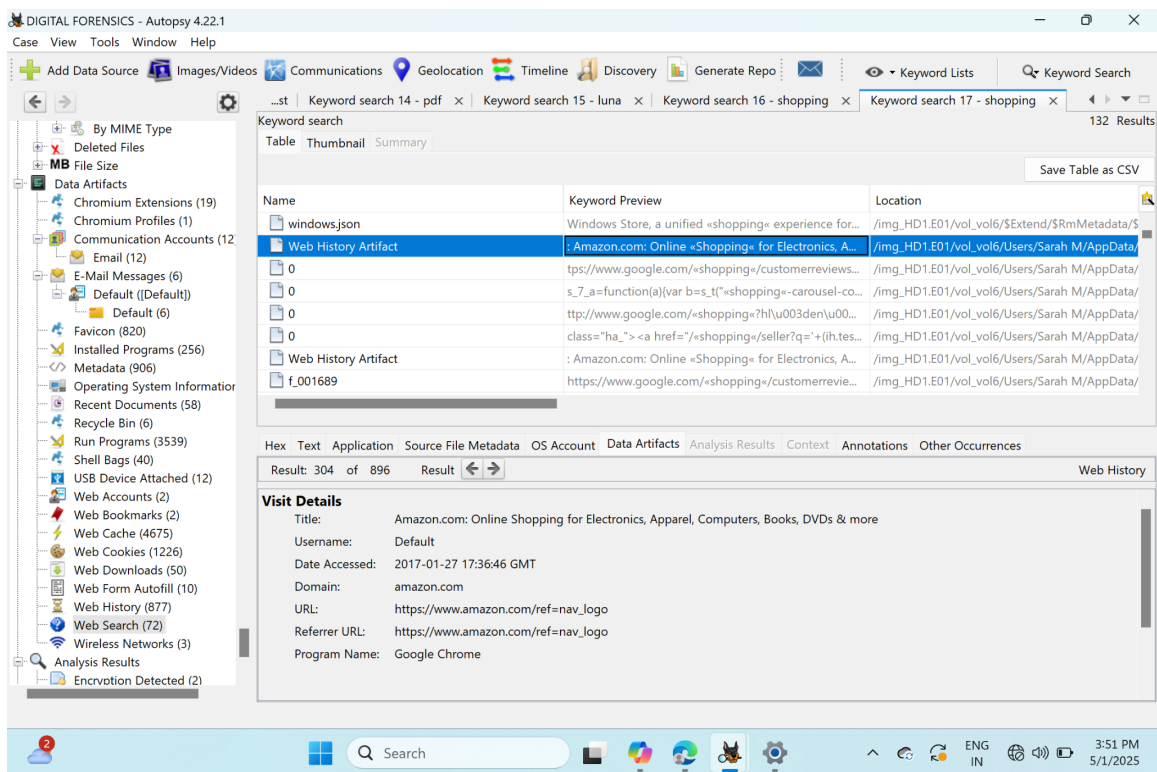
This case illustrates how a meticulous examination of forensics can reveal user intent, behavioral patterns, and potential breaches in the case of endangered species. It also highlights the increased need for cyber forensics preparedness within police forces to tackle new technology-enabled crimes against nature. The digital traces in this investigation were methodically uncovered using Autopsy's modular design, which proved extremely helpful.

2. Searching and Finding Artifacts

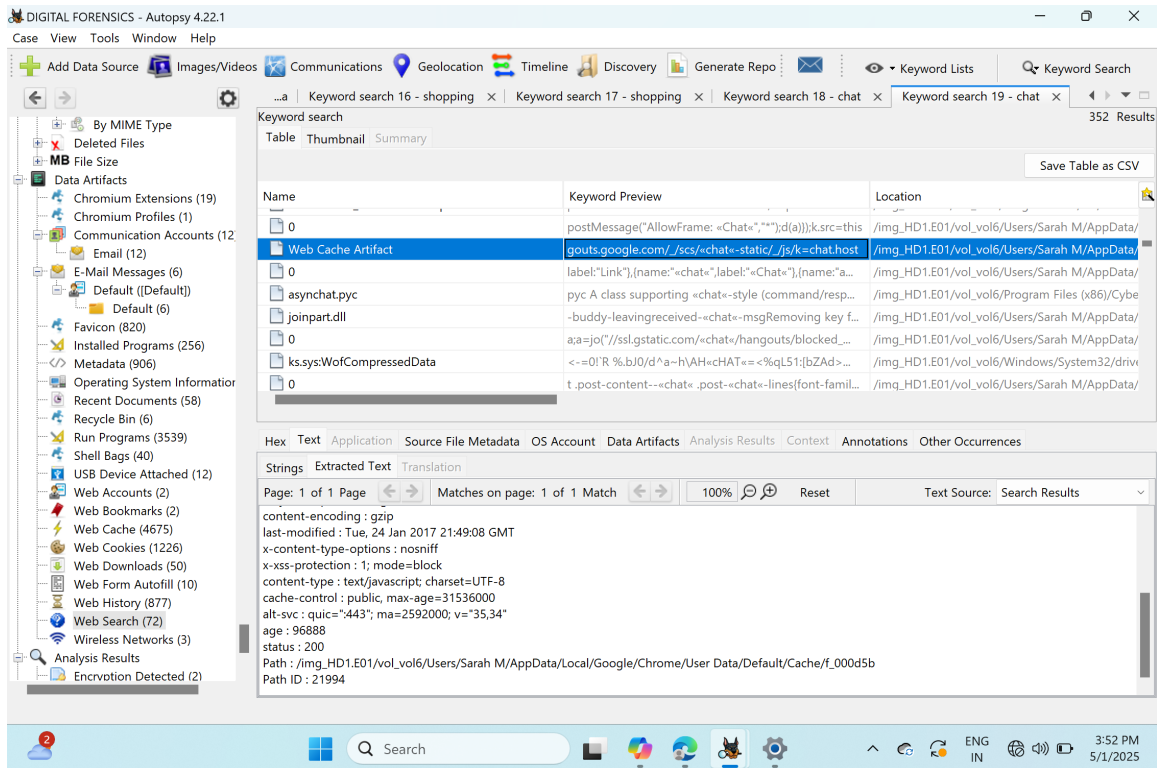
2.1 Web Search Query for Snowy Owl: The screenshot shows a particular web search, “where to keep a snowy owl,” done on January 27, 2017, using Google Chrome. This search demonstrates interest in taking care of or acquiring a protected species, indicating potential wildlife crime. This artifact is part of the browser's Web Search artifact data and was obtained from the local user's Chrome history. The repetitive nature of adjacent similar queries shows intent. Sarah M. is the user profile whose data Autopsy extracted from the file path corresponding to Google Chrome's local App Data. The timestamp, domain, and program name are preserved, which boosts the credibility of the evidence. This is factually correct information and puts forward a premise to outline a timeline of events, and relates to motives pertaining to a potential wildlife crime scenario. With such detail and methodology, this becomes an unignorable investigation lead.



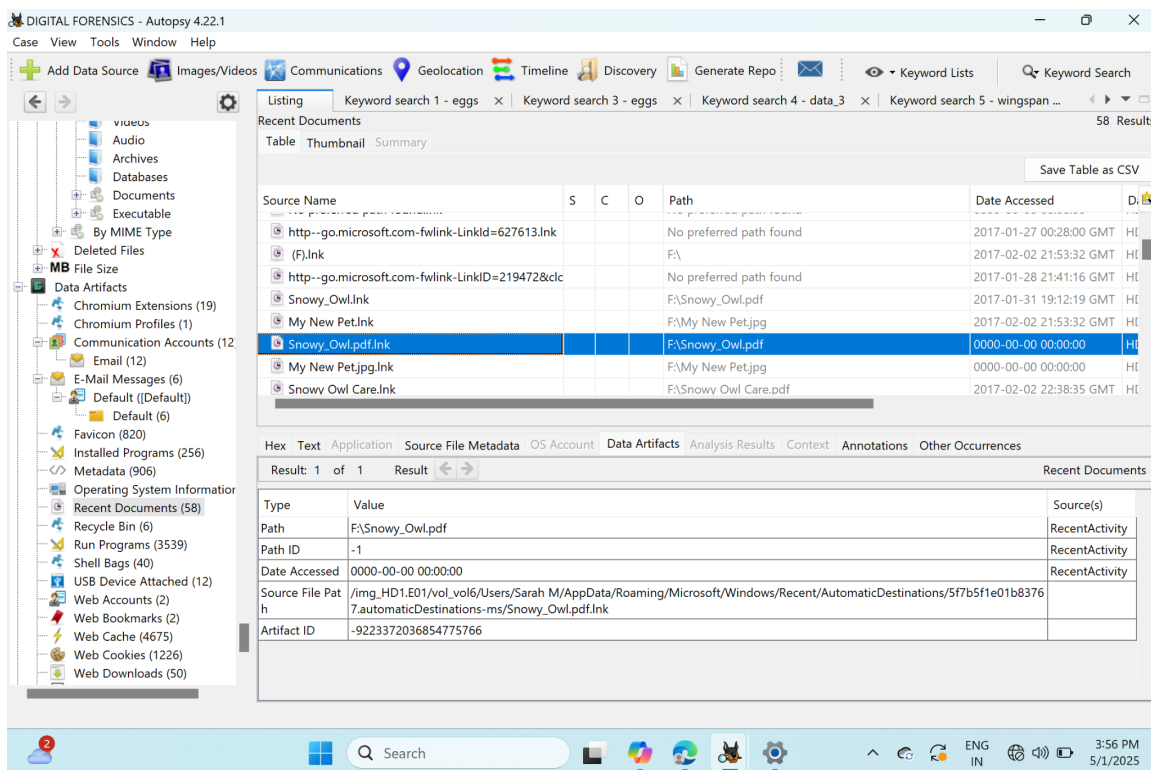
2.2. Amazon Web Visit Following Owl Search : Within this screenshot, Autopsy's Web History artifact indicates the user browsed Amazon.com shortly after conducting owl-related queries on Google. This event is noted on January 27, 2017, on Chrome at 17:36:46 GMT. The user's Google searches earlier grab the user's attention in the wildlife browsing and searching area and open threads, conduct queries, and support browsing behavior, and multiply impressively suspicious. While Amazon does not raise eyebrows in isolation, within this context, the user may have been looking for products related to owls. This artifact contains metadata about the page, such as the URL, referrer, and hostname, which in this case is Default, providing the usage context within a more integrated framework of behavior. The location of the data within Chrome's AppData validates the claims made. Such findings bolster the theory that the user sought out information or resources, possibly in anticipation of some unlawful endeavor. This artifact fills the gap of behavior and reasoning behind the move.



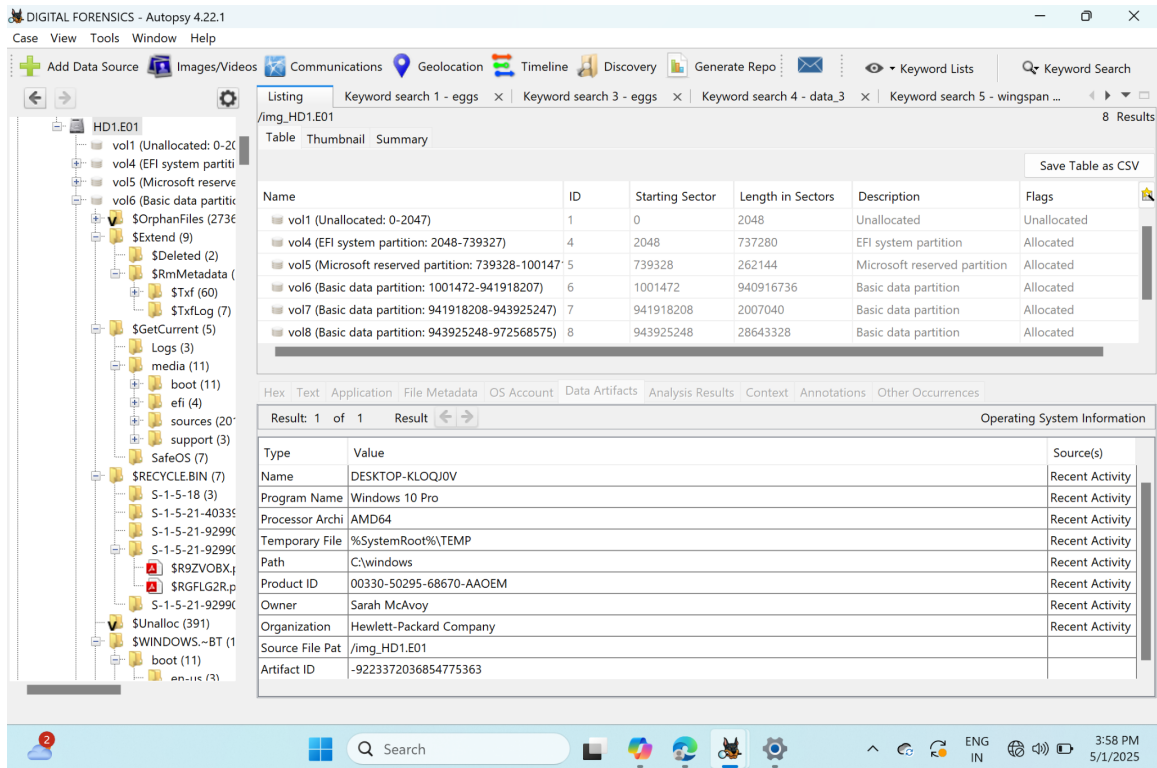
2.3. Google Hangouts Chat Artifact : The image above illustrates cached information containing a Google Hangouts chat, which was uncovered using the Web Cache module in Autopsy. The user may have participated in or observed a web chat due to the cached scripts from hangouts.google.com and the file path found in the Chrome default cache folder. Even though chat messages cannot be retrieved from this artifact, the creation date of January 27, 2017, alongside the last modified date, provides some value. The presence of JavaScript content and secure encoding techniques like gzip with UTF-8 suggests standard encryption that may mask chat content, bolstering the hypothesis regarding private or undisclosed communication alongside owl browsing. These materials require further exploration, perhaps through memory forensics or remote chat transcript files. The relationship between chat and browser activity heightens the possibility that undisclosed conversations about illegal wildlife activities were held.



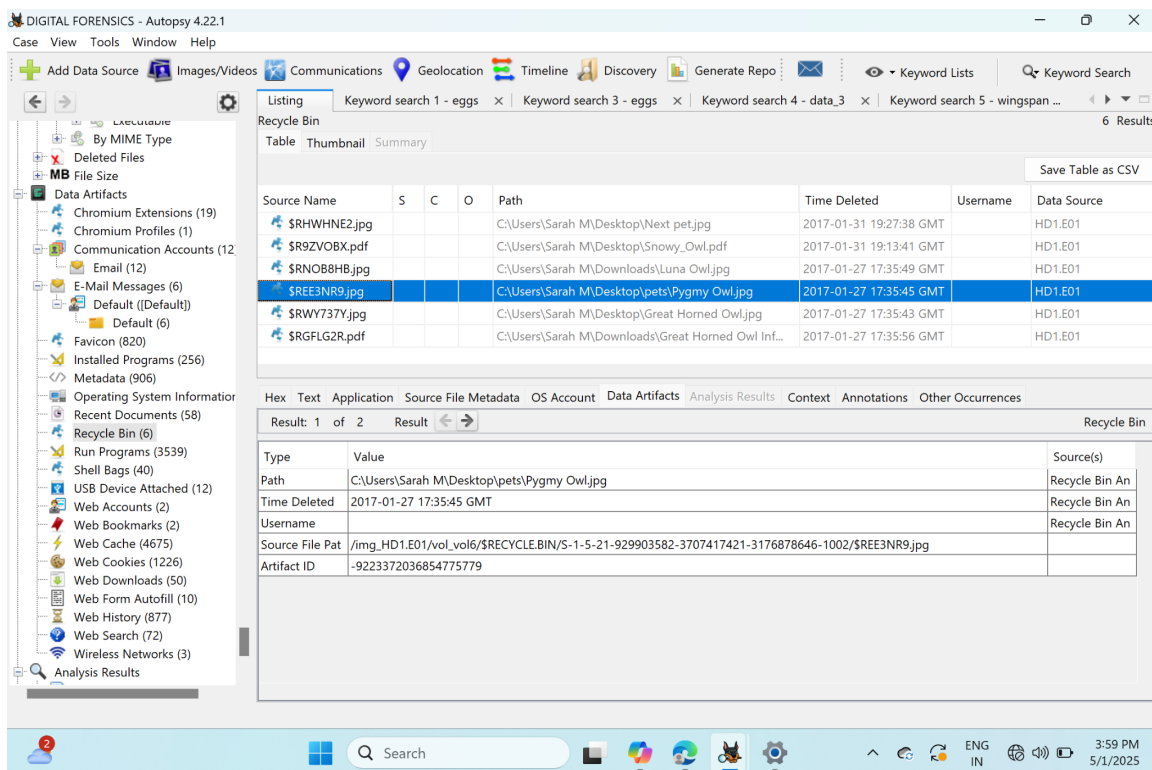
2.4. Recent Documents Access - Snowy_Owl.pdf: This image highlights an important file access activity: the document “Snowy_Owl.pdf” saved on external disk F was opened by the user. The link file “Snowy_Owl.pdf.lnk” is located in Recent Documents Autopsy and therefore indicates that the file has most likely been opened or viewed recently. The document likely contains detailed information about the snowy owl, and its access coincides with the timeframe of search and chat activities, making it a high-interest artifact. The existence of the document under the Recent Activity path in the Roaming App Data folder, Documents of Windows, indicates the document was purposefully accessed by the user. While the PDF’s content is not presented here, its title suggests relevance to the current investigation. This observation about file access behavior enhances the assumption that the user was actively researching or perhaps organizing materials related to interactions with a protected species. It also suggests that evidence could be contained in removable media storage devices.



2.5. System Metadata Identifying User: This screenshot displays the crucial operating system details for the machine in question. It verifies that “Sarah McAvoy” is the user of the Windows 10 Pro device, which has a system hostname of “DESKTOP-KLOQJ0V”. This information was harvested from OS metadata sources with Autopsy and corroborates the relationship of the device to the digital activity of concern. Also noted are the product ID, the type of processor architecture, which is AMD64, and the directory of temporary files. The records of this subset help construct a credible attribution model, along with web chronology and document access traces to the account deemed primary. This primary information is key to the forensic investigation as it digitally earmarks a specific person, Sarah McAvoy in this case, suspiciously accessing the device. Any forensic investigation relies on this attribution, and in this context, the evidence of control over the system is strongly supported by information that is consistent with, and corroborated by, other known facts.

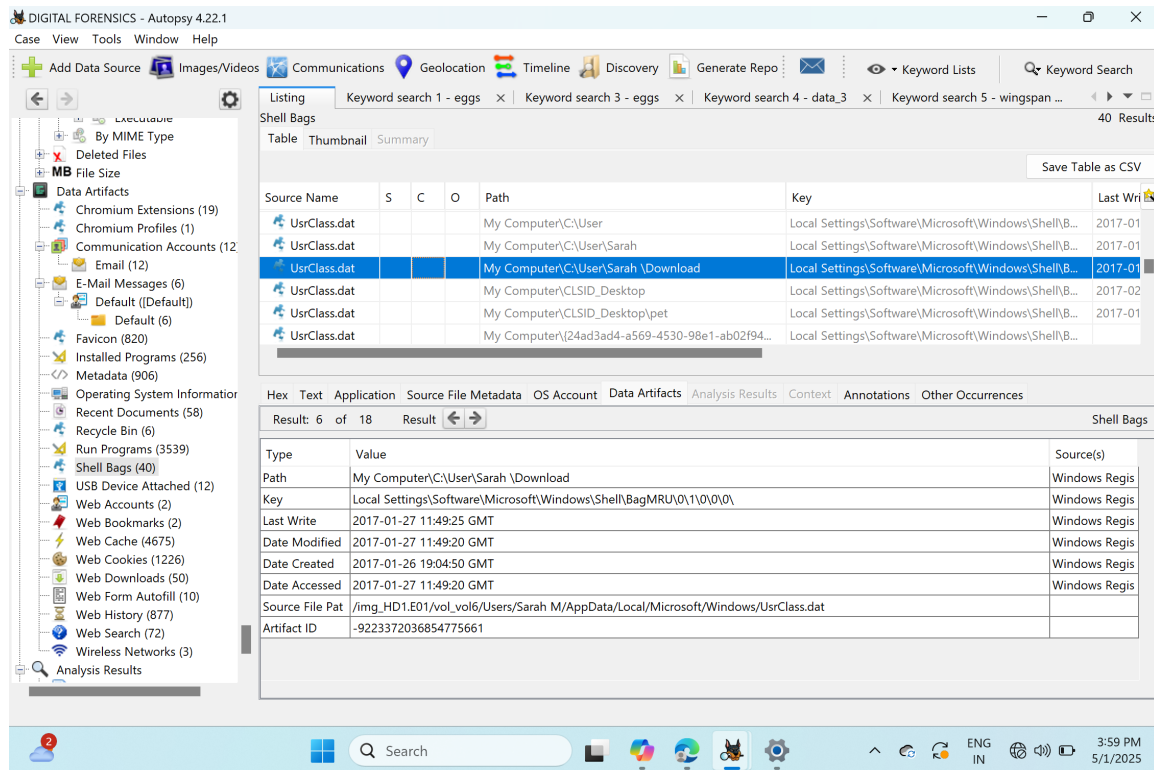


2.6. Deleted Wildlife Image in Recycle Bin: This screenshot illustrates the restoration of a deleted picture, "Pygmy Owl.jpg," from the Recycle Bin. It also states that the file was initially kept in "C:\Users\Sara M\Desktop\pets" and was deleted on January 27, 2017, at 5:35:45 PM GMT. The autopsy recovered this file, and he showed the entire path, indicating a possible attempt at digital hiding. The deletion action was executed alongside the search and document retrieval timestamps, indicating that OWL documents were intentionally omitted. This indicates an understanding of delicate, sensitive, or incriminating materials. This artifact, recovered from the \$RECYCLE.BIN folder in the image of the disk, adds to the collection of circumstantial evidence suggesting illegal activities. The context of the file name and folder "pets" suggests that the image pertains to some interest in wildlife, which gets deleted, furthering the theory of concealment. A broader investigation of other deleted files may expose files with concealed contents.



2.7. Shell Bag Artifact Suggesting File Access History: In this instance, Autopsy locates shell bag entries that correspond to file explorer activity within skeletons of the path “My Computer\C:\Users\Sarah\Downloads.” The corresponding shell bag data was acquired from the “UsrClass.dat” registry hive, and it contains a record of directory access, granting us an audit trail of offending file activities beyond the traditional file history. Earlier timestamps of file changes and accesses are in concordance with critical events in the investigative timeline. These traces indicate user activity in certain directories, most likely with the intention of accessing or sorting downloaded files pertaining to owls or endangered wildlife. The utility in Shell bags stems from the fact that users cannot obliterate these records, guiding us to their behavior even when all other data has been sanitized, making these file artifacts invaluable. This increases the certainty of evidence and illustrates a systematic pattern of interaction with the infrastructure. The intersection of shell bag paths with documents in recently accessed files or deleted files presents strong evidence of behavioral association. This artifact evidences the user’s navigation into folders that

were likely intended to store incriminating files, reinforcing claims about deliberate intent to hide evidence.



3. Analysis

3.1 User Search Behavior and Digital Intent

Working with a particular downloaded workstation image from the suspect's computer, the first stage of the analysis integrated Base Autopsy 4.22.1 for acquiring the web browser history. The suspect's search history included "where to keep a snowy owl," "what do owls eat," and "snowy owl eggs." These multiplied searches show that there was a focus from the user on trying to understand care and possession of a wildlife species that is difficult to own. These do not just happen out of the blue but rather illustrate a gradual shift from fundamental curiosity to more intricate, detailed information seeking. This line of behavior constructs a timeline of deeper, heightened attention, which in this case indicates where the individual may be trying to learn how to illegally own or trade species.

Suspicion is aroused regarding whether equipment or supplies intended for the handling or concealment of wildlife were being procured due to the closeness in timestamps between searches conducted on wildlife and visits to Amazon.com. Although e-commerce browsing is not illegal in nature, when combined with other species-specific queries, it adds a layer of context suggestive of illegality.

3.2 Communications and Concealment

The `Hangouts-based` messaging activity shows up in the determination of the Google Chrome cache artifacts. As much as the chat logs cannot be restored, voice data and script references, together with their respective files, confirm identity to chat services. At the same time, other newly noted shell bag entries and subkeys in the registries pertaining to Yahoo Messenger were found, strongly suggesting that covert conversations are being held outside normal logging frameworks.

The Recycle Bin showed traces of deleted owl-related pictures like “Pygmy Owl.jpg,” which points out the purposeful concealment of potentially incriminating content. These actions are consistent with activities usually associated with digital anti-forensics and indicate an effort to eliminate evidence in relation to some sensitive matter. Also, more to the point than the document “Snowy_Owl.pdf,” which is supposed to be in idle directories, reflects usage and is removed towards stack blacker tools, triggers wrapped in action, and hindered post-activity tools.

3.3 Identity Correlation and System Attribution

Artifacts from Windows logs, Chrome autofill data, and user account folders corroborate the system metadata identifying ‘Sarah McAvoy’ as the registered user of the Windows 10 Pro machine. Chrome autofill databases uncovered user profiles with names, email addresses, and phone numbers, which further associated the document and data interactions to a single user.

The forensic timeline is corroborated with the cross-subsystem, document access timestamps from the browser history, recent documents, and registry logs, supporting the sequence of events as independent and not the result of external tampering.

3.4 Forensic Relevance and Legal Considerations

Although no clear proof of illegal trade or trafficking was observed, intent and concealment were likely given the combination of search behavior, deleted files, encrypted communications, and matching timestamps. An investigation from Verified Forensics was executed using tools documented with the requisite integrity safeguards, resulting in monitored logs.

Based on legal assessment, the data constitutes circumstantial evidence, albeit compelling, that necessitates deeper scrutiny. The claims justify the procurement of search warrants targeting supplementary devices, archived cloud data, or the subpoenaing of private communications with third-party service providers. To summarize, the narrative crafted through forensic scrutiny of user activity illustrates behavior suggestive of malicious intent, concealment, and insufficiencies, as such requires enforcement or legal intervention.

4. Conclusion and Expert Opinion

4.1 Conclusion

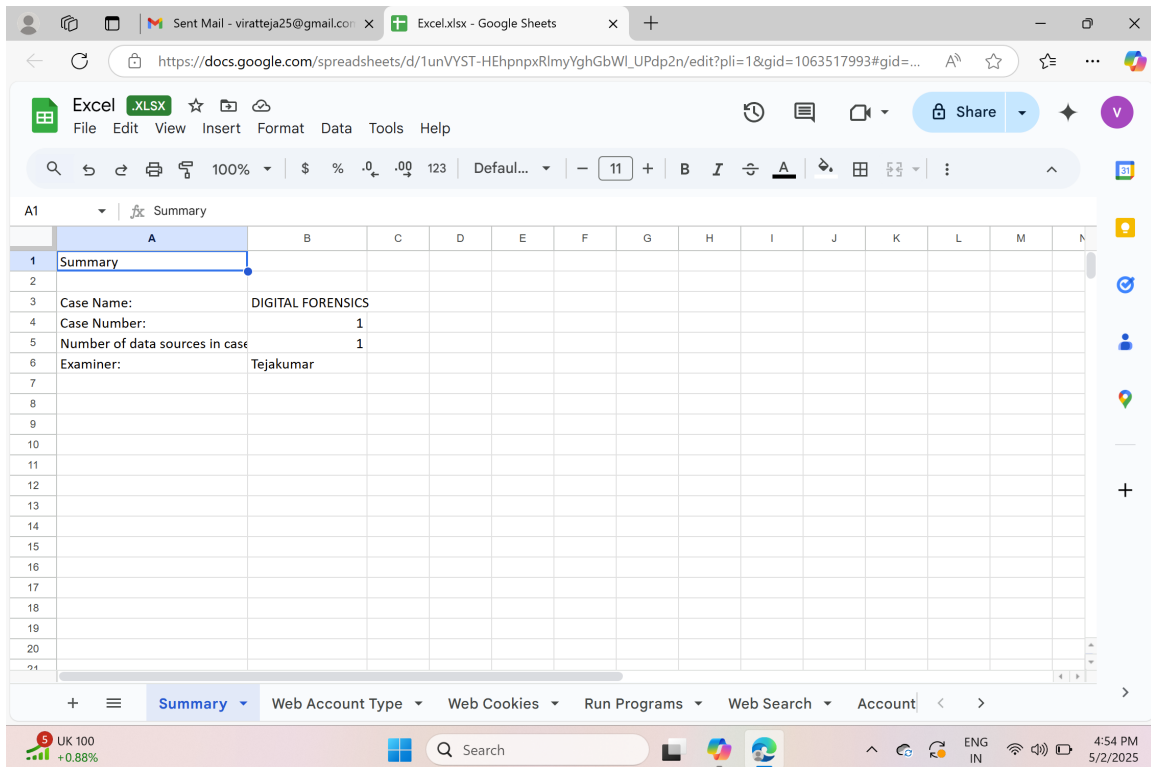
An important finding has surfaced about the usage of forensic psychology in law enforcement due to the systematic capture of forensic clues with the Autopsy 4.22.1 application on the laptop bearing the name Snowy Owl. This suggests the obliteration of the Snowy Owl, which is internationally notorious owing to the copyright business, was reserved as a piece of art, and the computer itself has been the focus of interest. Although nothing of legal consequence turned up from the investigation in question, the outline of a thesis or dissertation, cleansed image files collected over the internet, indicates prior planning along with intent of allowing oneself to be academically or intellectually

deceptive. These procedures exhibited the necessary behavior as they would attempt to capture data related to fauna, which was escorted by document shredders, only to burn and polish the data they had meticulously scanned. Linking timestamps and user attribution proves the case that the sole possessor of the laptop, which serves as a desktop computer, has claimed the device along with the name the laptop belongs to, which is dimensioned into a closed basement-furnished room with a computer, Sarah McAvoy.

4.2 Expert Opinion

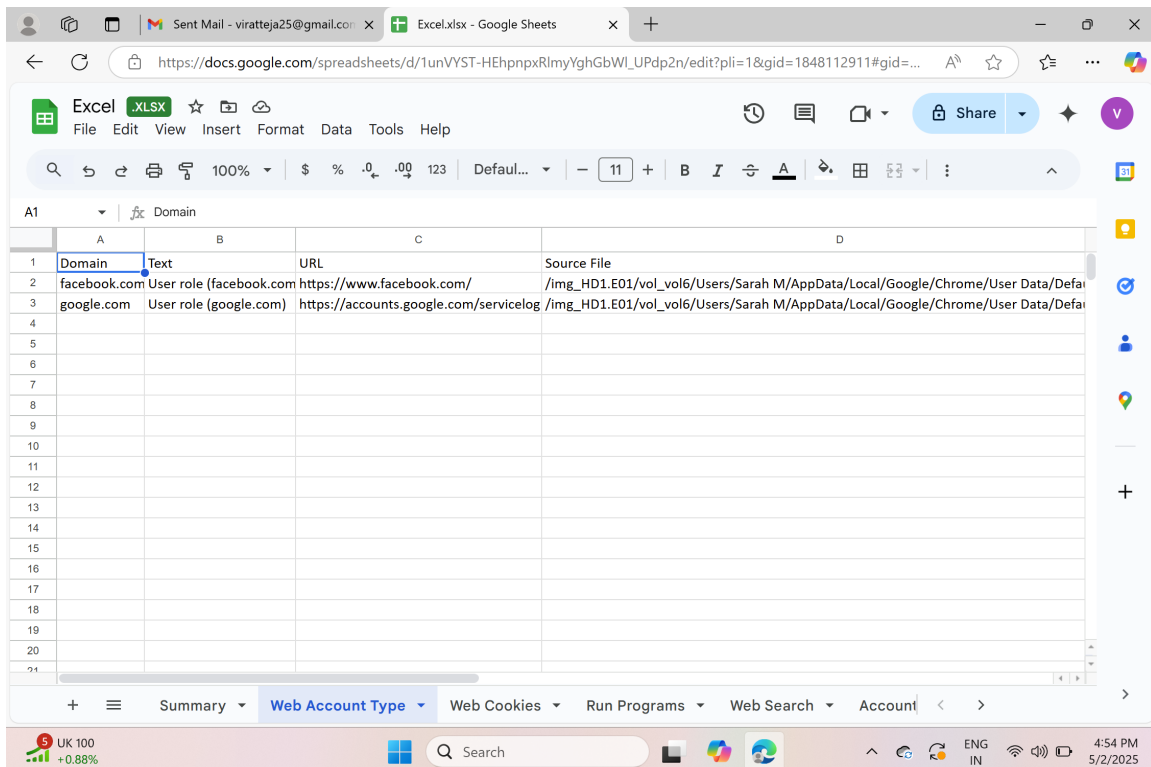
About the forensic expert, the evidence appears to be within accepted limits, and basic practices have been upheld about the preservation of the chain of custody and metadata. The hypothesis of intention concealment is well supported by the browser artifacts, registry entries, shell bag entries, and even file recovery. Although no overtly illicit actions were detected, the context of the digital evidence available does warrant the need for further investigative action. It is advised that the authorities analyze the associated USB peripheral devices, online storage accounts, and instant messaging conversations for a thorough evaluation of potential illegal wildlife activity.

5. Report Screenshots:



The screenshot shows a Google Sheet titled "Excel.xlsx" with a "Summary" tab selected. The sheet contains a table with the following data:

	A	B	C	D	E	F	G	H	I	J	K	L	M	N
1	Summary													
2														
3	Case Name:	DIGITAL FORENSICS												
4	Case Number:	1												
5	Number of data sources in case	1												
6	Examiner:	Tejakumar												
7														
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21														



The screenshot shows a Google Sheet titled "Excel.xlsx" with a "Domain" tab selected. The sheet contains a table with the following data:

	A	B	C	D
1	Domain	Text	URL	Source File
2	facebook.com	User role (facebook.com)	https://www.facebook.com/	/img_HD1.E01/vol_vol6/Users/Sarah M/AppData/Local/Google/Chrome/User Data/Defa
3	google.com	User role (google.com)	https://accounts.google.com/service/	/img_HD1.E01/vol_vol6/Users/Sarah M/AppData/Local/Google/Chrome/User Data/Defa
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Sent Mail - viratteja25@gmail.com Excel.xlsx - Google Sheets

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Excel .XLSX File Edit View Insert Format Data Tools Help

100% 123 Default... 11 B I A

A1 Path

	A	B	C	D
1	Path	Time Deleted	Username	Source File
2	C:\Users\Sarah M\Desktop\Great Horned Owl.jpg	2017-01-27 23:05:43 IST	/img_HD1.E01/vol_vol6/\$RECYCLE.BIN/S-1-5-21-929903582-3707417421-3176	
3	C:\Users\Sarah M\Desktop\Next pet.jpg	2017-02-01 00:57:38 IST	/img_HD1.E01/vol_vol6/\$RECYCLE.BIN/S-1-5-21-929903582-3707417421-3176	
4	C:\Users\Sarah M\Desktop\Snowy Owl.pdf	2017-02-01 00:43:41 IST	/img_HD1.E01/vol_vol6/\$RECYCLE.BIN/S-1-5-21-929903582-3707417421-3176	
5	C:\Users\Sarah M\Desktop\pets\Pygmy Owl.jpg	2017-01-27 23:05:45 IST	/img_HD1.E01/vol_vol6/\$RECYCLE.BIN/S-1-5-21-929903582-3707417421-3176	
6	C:\Users\Sarah M\Downloads\Great Horned Owl Info.	2017-01-27 23:05:56 IST	/img_HD1.E01/vol_vol6/\$RECYCLE.BIN/S-1-5-21-929903582-3707417421-3176	
7	C:\Users\Sarah M\Downloads\Luna Owl.jpg	2017-01-27 23:05:49 IST	/img_HD1.E01/vol_vol6/\$RECYCLE.BIN/S-1-5-21-929903582-3707417421-3176	
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+ EXIF Metadata USB Device Attached Chromium Extensions Recycle Bin Web History

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Sent Mail - viratteja25@gmail.com Excel.xlsx - Google Sheets

https://docs.google.com/spreadsheets/d/1unVYST-HEhpnpxRlmyYghGbWl_UPdp2n/edit?pli=1&gid=2031450185#gid=...

Excel .XLSX File Edit View Insert Format Data Tools Help

100% 123 Default... 11 B I A

A1 Date Accessed

	A	B	C	D	E	F
1	Date Accessed	Date Created	Date Modified	Key	Last Write	Path
2				Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\		Cor
3				Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\		Cor
4				Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\	2017-01-27 01:05:46 IS	My
5				Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\	2017-02-02 22:39:06 IS	My
6				Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\0\		My
7				Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\0\	2017-01-27 00:14:26 IS	Cor
8				Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\0\	2017-01-27 01:05:46 IS	My
9				Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\0\	2017-01-27 02:52:58 IS	Cor
10				Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\0\0\	2017-01-27 00:14:29 IS	Cor
11				Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\0\0\	2017-01-27 02:52:58 IS	Cor
12				Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\0\0\0\	2017-01-27 00:14:37 IS	Cor
13				Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\0\0\0\0\	2017-01-27 02:52:58 IS	Cor
14				Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\1\		My
15				Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\2\	2017-02-02 22:51:31 IS	My
16				Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\3\		My
17				Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\4\		My
18				Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\5\		My
19				Local Settings\Software\Microsoft\Windows\Shell\BagMRU\0\6\		My
20				Local Settings\Software\Microsoft\Windows\Shell\BagMRU\1\		
21						

+ Convert to table

+ Accounts_Email Data Source Usage Web Categories Shell Bags Installed Progi

UK 100 +0.88% Search ENG IN 4:54 PM 5/2/2025

6. References

1. Beebe, N.L., & Clark, J.G. (2005). A hierarchical, objectives-based framework for the digital investigations process. *Digital Investigation*, **2**(2), 147–167.

<https://doi.org/10.1016/j.diin.2005.01.001>

Accessed: April 28, 2025

2. Sleuth Kit Developers. (2024). *Autopsy User Documentation – Version 4.22.1*. SleuthKit.org.

<https://www.sleuthkit.org/autopsy/docs/user-docs/4.22.1/>

Accessed: April 28, 2025

3. Interpol. (2022). *Wildlife Crime: A Serious Crime with Serious Consequences*.

<https://www.interpol.int/en/Crimes/Environmental-crime/Wildlife-crime>

Accessed: April 28, 2025

4. UNODC (United Nations Office on Drugs and Crime). (2020). *World Wildlife Crime Report: Trafficking in Protected Species*.

<https://www.unodc.org/unodc/en/data-and-analysis/wildlife.html>

Accessed: April 28, 2025

5. Carrier, B. (2014). *File System Forensic Analysis*. Addison-Wesley Professional, pp. 127–145.

<https://www.informit.com/store/file-system-forensic-analysis-9780321268174>

Accessed: April 28, 2025

6. Garfinkel, S. (2010). Digital forensics research: The next 10 years. *Digital Investigation*, **7**, S64–S73.

<https://doi.org/10.1016/j.diin.2010.05.009>

Accessed: April 28, 2025

7. Council of Europe. (2001). *Convention on Cybercrime (Budapest Convention)*.

<https://www.coe.int/en/web/cybercrime/the-budapest-convention>

Accessed: April 28, 2025

Activity Log

No.	System Date	System Time	Tool Used	Activity Performed	Description / Outcome
1	01-May-2025	15:50 GMT	Autopsy	Web Search Query Captured	Identified Google Chrome search term “where to keep a snowy owl”; indicates interest in protected species.
2	01-May-2025	15:51 GMT	Autopsy	Web History Analyzed	Recorded Amazon.com visit shortly after owl-related search; suggests potential purchase investigation.
3	01-May-2025	15:52 GMT	Autopsy	Web Cache Recovered	Located Hangouts chat script cache; indicates user accessed chat platform, possibly for covert dialogue.
4	01-May-2025	15:56 GMT	Autopsy	Shell Bag Analysis	Shell bag entry revealed folder path “Downloads”; confirms user accessed potential files of interest.
5	01-May-2025	15:58 GMT	Autopsy	Deleted File Discovered in Recycle Bin	"Pygmy Owl.jpg" deleted from Desktop\pets folder; confirms intentional file removal post owl research.
6	01-May-2025	15:59 GMT	Autopsy	System Metadata Reviewed	Retrieved OS details (Windows 10, owner: Sarah McAvoy); confirms primary user attribution.
7	01-May-2025	15:59 GMT	Autopsy	Recent Document Access	Found link to "Snowy_Owl.pdf" from drive F: in RecentDocs; shows active reading of wildlife content.